happens can aid troubleshooting. At the very least, it should help you to determine if the problem is on the provider side or in your app.

We can differentiate between sources you own—and, hence, you're free to instrument—and sources you don't own. Typically, the sources you don't own—and, hence, aren't free to instrument—make up the majority of the sources. This ranges from the compute runtime to storage and network to libraries and other services you use and depend on in your own code.

OK, that was enough theory. Now, let's get our hands dirty, starting with compute-related sources.

3.2 Compute-related sources

This section reviews compute-related sources. We will have a look at different types of compute, their main characteristics from an observability angle, and what signals are relevant for which roles.

3.2.1 Basics

In the context of cloud-native environments, we're usually faced with a variety of options when it comes to compute. Ultimately, we care about the features we can deliver to business stakeholders. So in an ideal world, we would be able to focus on the application level, which is core to the business, and outsource as much of the heavy lifting (e.g., security patches and operating system upgrades, which are typically not core to the business) to our service provider of choice.

There are oftentimes, however, good reasons that we want to or must own a larger part of the infrastructure stack than we may desire—from compliance requirements to the need for controlling certain aspects of compute (like GPU- or I/O-related). No matter where exactly the line between the infrastructure and application level is, in your case, a good understanding of the responsibilities (e.g., who owns upgrades or security aspects) is the basis for a successful cloud-native operation.

Depending on your role (operations, developer, architect, release engineer, etc.), you might view compute in different ways. Some common ones include the following:

- The unit of deployment (a VM, container, or function)
- Server-centric versus serverless compute
- Monolithic apps versus service-oriented (microservices)

Figure 3.1 shows the spectrum of compute with some common cloud-native example setups. The unshaded boxes in each setup stand for the respective infrastructure components and the shaded boxes for the application-level components.

> **NOTE** In reality, in any given production environment, you will come across a combination of compute. This may be due to technical reasons (e.g., a move to the cloud and, with it, a containerization of the service), or it could be due to business reasons, such as an acquisition or merger, where different architectures need to be integrated. No matter the reason, it means you almost